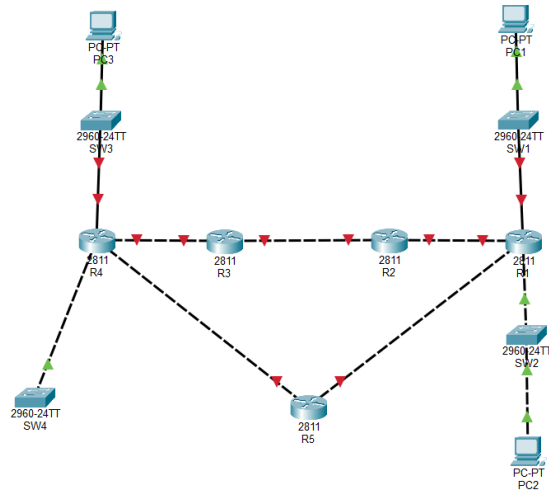


Week 7: Routing Fundamentals (10 Marks)

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52) R1

```
Router>ena
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R1
R1(config)#no shutdown
```

R2

```
Router>ena
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R2
R2(config)#exit
R2#
%SYS-5-CONFIG_I: Configured from console by console
```

R3

```
Router>ena
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R3
R3(config)#exit
R3#
%SYS-5-CONFIG_I: Configured from console by console
```

R4

```
Router>ena
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R4
R4(config)#exit
R4#
%SYS-5-CONFIG_I: Configured from console by console
```

53)

```
Router>ena
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface FastEthernet
Router(config)#
Router(config)#interface FastEthernet0/0
Router(config-if)#ip address 10.0.1.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

Router(config-if)#interface FastEthernet 0/1
Router(config-if)#ip address 10.0.2.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
```

54)

```
Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.2.0/24 is directly connected, FastEthernet0/1
L       10.0.2.1/32 is directly connected, FastEthernet0/1

Router#
```

55)

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.0.2.10

Pinging 10.0.2.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.0.2.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>tracert 10.0.2.10

Tracing route to 10.0.2.10 over a maximum of 30 hops:
  0  *         *         *         Request timed out.
  1  *         *         *         Request timed out.
  2  *         *         *         Request timed out.
  3  *         *         *         Request timed out.
  4  *         *         *         Request timed out.
  5  *         *         *         Request timed out.
  6  *         *         *         Request timed out.
  7  *         *         *         Request timed out.
  8  *         *         *         Request timed out.
  9  *         *         *         Request timed out.
 10 *         *         *         Request timed out.
 11 *         *         *         Request timed out.
 12 *         *         *         Request timed out.
 13 *         *         *         Request timed out.
 14 *         *         *         Request timed out.
 15 *         *         *         Request timed out.
 16 *         *         *         Request timed out.
 17 *         *         *         Request timed out.
 18 *         *         *         Request timed out.
 19 *         *         *         Request timed out.
 20 *         *         *         Request timed out.
 21 *         *         *         Request timed out.
 22 *         *         *         Request timed out.
 23 *         *         *         Request timed out.
 24 *         *         *         Request timed out.
 25 *         *         *         Request timed out.
 26 *         *         *         Request timed out.
 27 *         *         *         Request timed out.
 28 *         *         *         Request timed out.
 29 *         *         *         Request timed out.
 30 *         *         *         Request timed out.

Trace complete.
```

56)

```
C:\>ping 10.1.2.10
Ping request could not find host 10.1.2.10. Please check the name and try again.
C:\>
```

57) R2

```
Router>ena
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#int f0/0
Router(config-if)#ip address 10.0.0.2 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#int f0/1
Router(config-if)#ip address 10.1.0.2 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
```

R3

```
Router>ena
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int f0/0
Router(config-if)#ip address 10.1.0.1 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

Router(config-if)#int f0/1
Router(config-if)#ip address 10.1.1.2 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
```

R4

```
Router>ena
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int f0/0
Router(config-if)#ip address 10.1.1.1 255.255.255.0
Router(config-if)#no sht
^
% Invalid input detected at '^' marker.

Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#int f0/1
Router(config-if)#ip address 10.1.2.1 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
```

58)

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.1.2.1

Pinging 10.1.2.1 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.1.2.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

59)

```
Router>ena
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip route 10.1.0.0 255.255.255.0 10.0.0.1
Router(config)#ip route 10.0.2.0 255.255.255.0 10.0.0.1
^
% Invalid input detected at '^' marker.

Router(config)#ip route 10.0.2.0 255.255.255.0 10.0.0.1
Router(config)#ip route 10.0.3.0 255.255.255.0 10.0.0.1
Router(config)#ip route 10.1.1.0 255.255.255.0 10.1.0.1
Router(config)#ip route 10.1.2.0 255.255.255.0 10.1.0.1
Router(config)#ip route 10.1.3.0 255.255.255.0 10.1.0.1
Router(config)#S
```

60)

```
C:\>ping 10,1.2.10
Ping request could not find host 10,1.2.10. Please check the name and try again.
C:\>ping 10.1.2.10

Pinging 10.1.2.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.1.2.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

61)

```
R1>ena
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#no ip route 10.1.0.0 255.255.255.0 10.0.0.2
R1(config)#no ip route 10.1.1.0 255.255.255.0 10.0.0.2
R1(config)#no ip route 10.1.2.0 255.255.255.0 10.0.0.2
R1(config)#no ip route 10.1.3.0 255.255.255.0 10.0.0.2
R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console
```

62)

```
%SYS-5-CONFIG_I: Configured from console by console

R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#ip route 10.1.0.0 255.255.255.0 10.0.0.2
R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console
```

63)

64)

```
Router>ena
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int f0/0
Router(config-if)#ip address 10.1.3.2 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#int f0/1
Router(config-if)#ip address 10.0.3.2 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
```

65)

```
C:\>ping 10.1.3.2

Pinging 10.1.3.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.1.3.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

66)

```
Router>ena
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#10.1.3.0 255.255.255.0 10.0.3.2
      ^
% Invalid input detected at '^' marker.

Router(config)#ip route 10.1.3.0 255.255.255.0 10.0.3.2
Router(config)#
```

67)

```
C:\>tracert 10.1.3.2
Tracing route to 10.1.3.2 over a maximum of 30 hops:
  0  *          *          *          Request timed out.
  1  *          *          *          Request timed out.
  2  *          *          *          Request timed out.
  3  *          *          *          Request timed out.
  4  *          *          *          Request timed out.
  5  *          *          *          Request timed out.
  6  *          *          *          Request timed out.
  7  *          *          *          Request timed out.
  8  *          *          *          Request timed out.
  9  *          *          *          Request timed out.
 10  *          *          *          Request timed out.
 11  *          *          *          Request timed out.
 12  *          *          *          Request timed out.
 13  *          *          *          Request timed out.
 14  *          *          *          Request timed out.
 15  *          *          *          Request timed out.
 16  *          *          *          Request timed out.
 17  *          *          *          Request timed out.
 18  *          *          *          Request timed out.
 19  *          *          *          Request timed out.
 20  *          *          *          Request timed out.
 21  *          *          *          Request timed out.
 22  *          *          *          Request timed out.
 23  *          *          *          Request timed out.
 24  *          *          *          Request timed out.
 25  *          *          *          Request timed out.
 26  *          *          *          Request timed out.
 27  *          *          *          Request timed out.
 28  *          *          *          Request timed out.
 29  *          *          *          Request timed out.
 30  *          *          *          Request timed out.

Trace complete.
```

```
Router>ena
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ping 10.0.1.10
      ^
% Invalid input detected at '^' marker.

Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#ping 10.0.1.10

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.1.10, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)

Router#
```

68)

```
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int f0/1
Router(config-if)#ip address 203.113.1 255.255.255.0
      ^
% Invalid input detected at '^' marker.

Router(config-if)#
```

69)